immediately as a 1 trip failure, it may take up to five minutes of accumulated trouble confirmation to set the DTC and illuminate the MIL. Three consecutive successful OBDII/EURO III trips or clearing the DTCs with a diagnostic tool (DRBIII® or equivalent) is required to extinguish the MIL. When the Transmission requests that the PCM illuminate the MIL, the PCM sets a DTC P0700 ($89) to alert the technician that there are DTCs in the Transmission. The PCM DTC ($89) must also be erased in the PCM in order to extinguish the MIL.

3.3.1 HARD CODE

POWERTRAIN

A diagnostic trouble code that comes back within one cycle of the ignition key is a ″hard″ code. This means that the problem is present when the PCM checks that circuit or function. Most procedures in this manual verify if the trouble code is a hard code at the beginning of each test. When it is not a hard code, an ″intermittent″ test must be performed.

Codes that are for OBDII monitors will not set with just the ignition key on. Comparing these to non-emission codes, they will seem like an intermittent. These codes require a set of parameters to be performed (The DRBIII® pre-test screens will help with this for MONITOR codes), this is called a ″TRIP″. All OBDII DTCs will be set after two or in some cases one trip failures, and the MIL will be turned on. These codes require three successful, no failures, TRIPS to extinguish the MIL, followed by 40 warm-up cycles to erase the code. For further explanation of TRIPS, Pre-test screens, Warm-up cycles, and the use of the DRBIII®, refer to the On Board Diagnostic training booklet #81-699-97094.

TRANSMISSION

Any Diagnostic Trouble Code (DTC) that is set whenever the system or component is monitored is a HARD code. This means that the problem is there every time the PCM checks that system or component. Some codes will set immediately at start up and others will require a road test under specific conditions. It must be determined if a code is repeatable (Hard) or intermittent before attempting transmission diagnosis.

3.3.2 ONE TRIP FAILURE

A One Trip Failure, when read from the PCM, is a hard OBDII/EURO III code that has not matured to the full 5 minutes. This DTC can take up to five minutes of problem identification before illuminating the MIL

3.3.3 INTERMITTENT CODE

A diagnostic trouble code that is not present every time the PCM checks the circuit is an ″intermittent'' code. Most intermittent codes are caused by wiring or connector problems. Intermittent conditions that come and go like this are the most difficult to diagnose; they must be looked for under specific conditions that cause them. The following procedures may assist you in identifying a possible intermittent problem:

- Visually inspect related wire harness connectors. Look for broken, bent, pushed out, or corroded terminals.
- Visually inspect the related harnesses. Look for chafed, pierced, or partially broken wire.
- Refer to any S.T.A.R. Hotline Newsletters, Service Information Tune ups (SITs) or Service Bulletins that may apply.
- Use the DRBIII® data recorder or co-pilot.

Some Transmission intermittent DTCs are caused by wiring or connector problems. However intermittent Speed ratio codes are usually caused by intermittent hydraulic seal leakage in the clutch and/or accumulator circuits. Intermittent speed ratio codes can be set by intermittent speed sensor circuitry or by line noise being induced onto one or both of the speed sensor signal circuits. Problems that come and go like this are the most difficult to diagnose, they must be looked for under the specific conditions that cause them.

3.3.4 STARTS SINCE SET COUNTER

POWERTRAIN

This reset counter counts the number of times the vehicle has been started since codes were last set or erased. This counter will count up to 255 start counts. The number of starts helps determine when the trouble code actually happened. This is recorded by the PCM and can be viewed on the DRBIII® as STARTS since set. When there are no trouble codes stored in memory, the DRBIII® will display ″NO TROUBLE CODES FOUND'' and the reset counter will show ″STARTS since set = XXX.'' OBDII vehicles will also display a DTC Specific or Global ″Good Trip″ counter which will indicate the number of ″Good Trips″ since the DTC was set. After 3 consecutive ″Good Trips,″ the MIL is extinguished and the good trip counter is replaced by a ″Warm Up Cycle″ counter. 40 Warm-Up Cycles will erase the DTC and Freeze Frame information.

TRANSMISSION

The Starts Since Set counter counts the number of times the vehicle has started since the most